Steering Large Language Models for Emotion Control: A Systematic Study of Prompts, Vectors, and Conceptors

Paarth Jindal
2023101050

Virat Garg
2023101081

Varun Gupta
2023101108

Abstract

Large Language Models (LLMs) excel at factual and semantic reasoning but struggle to convey and sustain emotional states in dialogue. This limitation hinders their use in empathetic interaction, education, and affective computing. Prior approaches to emotion-aware text generation rely on prompt engineering or fine-tuning, which are brittle and resource-intensive. Recent advances in activation steering enable lightweight, representation-level control, yet most existing work targets narrow attributes (e.g., single word translation, antonymy, capitalization) with limited evaluation in open-ended emotional response generation.

In this work, we systematically analyze emotional steering across two method families: (1) additive emotion vectors (EVs) extracted and, and (2) conceptor-based steering. Our study reveals that while additive EVs provide strong control over emotional aware outputs, vanilla conceptors fail to generalize due to activation bias and over-regularization for open ended tasks. Building on this diagnosis, we introduce CoRAL (Conceptor Residual Alignment Layer), a residual–affine reformulation that stabilizes conceptor steering without altering model semantics. Extensive ablations over layers, aperture scaling, and activation geometry (t-SNE) offer insight into controllability and failure modes. Together, these results establish a principled framework for analyzing and improving emotion aware steering in LLMs.

1 Introduction

Large Language Models (LLMs) achieve impressive results in reasoning and factual benchmarks, yet they often fail to maintain consistent emotional expression across text generation tasks. Outputs tend to drift toward neutral or generic tones, limiting their applicability in empathy-sensitive settings such as tutoring, counseling, and assistive communication (Bai et al., 2022). This challenge

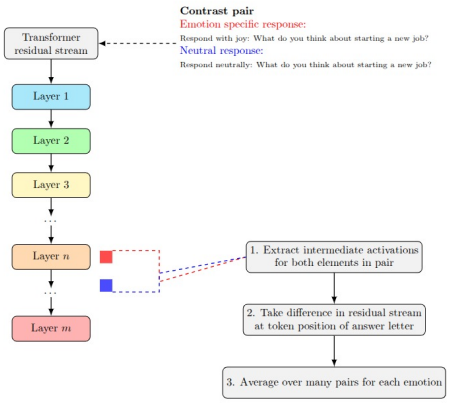


Figure 1: Steering Vector Generation

underscores the need for mechanisms that can modulate emotional style controllably and interpretably, without costly retraining.

Traditional approaches like prompt engineering and supervised fine-tuning only partially address this need. Prompts are brittle and highly context-dependent, while fine-tuning incurs high annotation and compute cost, scales poorly across emotional categories, and often harms generalization. To overcome these limitations, recent work explores activation-level steering, which manipulates a model's internal representations at inference time to guide behavior (Turner et al., 2023). Activation steering provides lightweight, model-agnostic control and forms the theoretical foundation of our study on emotion steering.

Within this paradigm, we focus on two main methods: (1) Additive Emotion Vectors (EVs)–mean-difference vectors extracted from emotionally tagged activations (Dong et al., 2025)–enable direct, linear modulation of affective style. We conduct extensive ablation studies on EVs, analyzing layer sensitivity, magnitude scaling, and geometric clustering (via t-SNE and spectral analysis) to understand how emotional subspaces form and interact. (2) Conceptor-based steering, introduced by

Jaeger (2014) and later adapted for transformers (Jørgensen et al., 2024), generalizes linear addition through covariance-aware "soft projections." Conceptors define ellipsoidal regions in activation space, allowing structured control and Boolean composition, but exhibit instability in emotional domains due to strong mean shifts and high intra-class variance.

Our analysis of these phenomena leads to CoRAL (Conceptor Residual Alignment Layer)–a residual–affine extension of conceptor steering that recenters activations and stabilizes projection while retaining semantic coherence. CoRAL unifies additive and conceptor-based approaches under a shared framework and supports interpretable, controllable emotional modulation in LLMs.

2 Background and Related Work

In this section, we review prior work on emotion-aware language generation, activation-based control, and conceptor steering, highlighting the strengths and limitations that motivate our proposed approach.

2.1 Emotion-Aware Text Generation

Controlling emotional expression in LLMs has been studied primarily through prompting and supervised fine-tuning. Controllable Emotion Generation by Dong et al. (2025) introduced the EmotionQuery and EmotionQuery+ benchmarks, enabling evaluation of discrete emotions using 100 prompt–response pairs per category. While effective for categorical affect control, these datasets focus on accuracy of discrete emotions rather than compositional or continuous affect spectra.

Madani et al. (2024) extended this idea to long emotional-support dialogues via fine-tuning LLaMA 2/3 models and proposed the Strategy-Relevant Attention metric to evaluate emotional consistency over multiple turns. However, their setup remains domain-specific and does not explore latent-space control or representation-level steering.

Similarly, Wang et al. (2024a) modeled affect as a stylistic attribute using style vectors, steering generative models toward sentiment or formality. Though conceptually adjacent, style vectors operate at a coarser granularity and cannot represent fine-grained emotional intensity or mixed emotions (e.g., frustration = sadness + anger).

To overcome the limited sample diversity of EmotionQuery+, we construct a synthetic extension, EmotionScape-Mini, using GPT-4o-mini with the same prompt template and parameters as Dong et al. (2025). This augmentation provides denser emotion coverage and supports our large-scale experiments on conceptor-based methods.

2.2 Activation Steering and Latent Interventions

A growing body of work manipulates hidden activations rather than model weights to control behavior at inference time.

Zhang et al. (2024) derive contrastive activation vectors to improve instruction following, demonstrating compositional control across constraints.

SADI (Wang et al., 2024b) introduces semantics-adaptive steering, adjusting vectors based on input context to reduce side effects from static steering directions.

Jorgensen and Albanie (2023) propose mean-centering of activations to mitigate global bias drift, improving stability in toxicity and genre control.

Despite these advances, most activation-steering methods remain purely additive, lacking mechanisms to adapt to covariance structure or residual alignment. Few studies examine how layer depth, steering magnitude, or representation geometry affect emotion control.

In this work, we extend additive Emotion Vectors (EVs) beyond fixed-vector operations by performing systematic ablation studies–analyzing layer sensitivity, scaling effects (λ), and latent-space geometry via t-SNE and spectral decomposition–to better understand the controllability and saturation of additive steering in emotional domains.

2.3 Conceptor Steering and Boolean Composition

Conceptor theory, introduced by Jaeger (2014), defines a soft projection operator derived from the covariance of internal states, allowing networks to recall and suppress patterns by projecting activations onto concept-specific subspaces.

The approach was later adapted to transformer activations by Jørgensen et al. (2024) in "Steering Large Language Models with Activation Deltas", where conceptors generalized linear steering vectors through affine projections and Boolean composition (AND/OR/NOT).

However, prior work on conceptor steering has only evaluated structured function tasks such as

antonymy, capitalization, translation, and plurality–tasks with clearly separable representations and deterministic outputs. No prior study has applied conceptor steering to behavioral, stylistic, or emotional open-ended generation, where activations exhibit strong mean shifts and high intra-class variance. As a result, conceptor steering's stability, interpretability, and controllability in affective domains remain untested.

We identify three key limitations of existing conceptor approaches:

- they assume zero-mean activations, which is invalid for emotion features;
- they ignore residual biases introduced by activation anisotropy; and
- they lack a mechanism to balance projection strength against semantic preservation.

2.4 Positioning of Our Work

Our work bridges the literature on emotion control (Dong et al., 2025) and activation-level steering (Zhang et al., 2024; Jorgensen and Albanie, 2023) with the theoretical foundation of conceptor projection (Jaeger, 2014; Jørgensen et al., 2024).

We show that classical conceptor steering collapses on affective tasks due to mean-shifted activations and propose CoRAL (Conceptor Residual Alignment Layer)–a residual–affine reformulation that recenters and stabilizes projection. CoRAL unifies additive Emotion Vectors and conceptor matrices within one framework, enabling continuous, interpretable control of emotional intensity.

Together with comprehensive ablations and geometric analyses, our study provides a principled methodological framework for diagnosing and improving emotion steering in large language models.

3 Proposed Methodology

Our goal is to develop structured and interpretable mechanisms for emotional steering in large language models (LLMs) through representation-space control. We focus on two complementary families of activation-level interventions: (1) Additive Emotion Vectors (EVs), representing first-order mean-shift steering, and (2) Conceptor-based steering, which introduces second-order, covariance-aware control. We further propose a new variant, CoRAL (Conceptor Residual Alignment Layer), that unifies both approaches under a single affine–residual formulation. Implementation details (datasets, models, and hyperparameters) are presented in section4.

3.1 Notation and Preliminaries

Let $h_\ell(x) \in \mathbb{R}^d$ denote the hidden activation of a model at layer ℓ for input x. We use the following notation consistently throughout:

Symbol	Definition
$\mathcal{E}_e, \mathcal{E}_0$	Emotion-conditioned and neutral prompt sets
$\mu_\ell^{(e)} = \mathbb{E}_{x\sim\mathcal{E}_e}[h_\ell(x)]$	Mean activation for emotion e
$\Sigma_\ell^{(e)} = \mathrm{Cov}_{x\sim\mathcal{E}_e}[h_\ell(x)]$	Covariance of activations
$v_\ell^{(e)} = \mu_\ell^{(e)} - \mu_\ell^{(0)}$	Additive emotion vector
$C_\ell^{(e)}$	Conceptor matrix at layer ℓ
α, γ	Aperture (regularization) parameters
λ	Steering strength at inference time
$b_\ell^{(e)}$	Bias term for mean correction in CoRAL

All activations and transformations are applied in $\mathbb{R}^d$. Throughout, expectation and covariance are computed over pooled token-level representations from the target layer.

3.2 Additive Emotion Vector (EV) Steering

3.2.1 Motivation

Additive steering has emerged as a simple yet powerful technique for controllable generation. Prior work showed that latent steering vectors, computed as differences in hidden activations between contrasting contexts, can be injected at inference to bias outputs toward desired behaviors such as safety, style, or emotion. Compared to prompt engineering, additive steering is less brittle and allows finer control at the representation level, but its performance depends critically on how the vectors are extracted and applied.

We adopt and extend this framework for emotional control, evaluating both baseline extraction methods and novel compositional variants.

3.2.2 Extraction of Emotion Vectors

We consider two extraction pipelines for an emotion e, both based on contrasting emotion-primed and neutral contexts:

Following Dong et al. (2025), we select a set of n neutral prompts $\{c_i\}_{i=1}^n$. For each, we collect hidden activations $h_\ell(c_i; e)$ when the model is prompted with emotion e, and $h_\ell(c_i; \text{neutral})$ for a

neutral variant. At layer ℓ, the emotion vector is:

$$v_e^{\text{prompt}} = \frac{1}{n}\sum_{i=1}^{n}\left(h_\ell(c_i; e) - h_\ell(c_i; \text{neutral})\right). \tag{1}$$

$$v_\ell^{\text{base}} = \frac{1}{|E|}\sum_{e\in E} v_e^{\text{ft}}, \tag{2}$$

v_ℓ^{base} denotes average EV vectora for all emotions and is used to get emotional aware outputs as compared to emotion specific outputs,denoted by v_e^{prompt}. This construction captures average shifts in representation space induced by the target emotion.

3.2.3 Injection of Emotion Vectors

At inference, the EV is added to hidden activations:

$$h'_\ell = h_\ell + \lambda \cdot v_e, \tag{3}$$

where λ is a tunable scalar controlling steering strength.

We test:

- **Single-layer injection:** adding at one fixed layer.
- **Multi-layer injection:** adding the same vector at multiple layers.
- **Scaling strategies:** following Anthropic's contrastive activation addition, we normalize vectors layer-wise and adjust magnitudes to prevent over-steering.



Figure 2: Emotion vector steering response examples

3.3 Conceptor Steering (Original Formulation)

Conceptor theory, introduced by Jaeger (2014), provides a principled framework for representing and manipulating subspaces of neural activations. A conceptor can be interpreted as a soft projection operator that captures the correlation structure of a specific activation manifold–allowing selective filtering and controlled modulation of latent representations. Conceptually, it behaves like a tunable linear filter that passes directions characteristic of a target concept while attenuating orthogonal components.

For an emotion-conditioned activation distribution $\mathcal{E}_e$ at layer ℓ, with empirical covariance $\Sigma_\ell^{(e)} = \text{Cov}_{x\sim\mathcal{E}_e}[h_\ell(x)]$, the corresponding conceptor matrix is defined as

$$C_\ell^{(e)} = \Sigma_\ell^{(e)}\left(\Sigma_\ell^{(e)} + \alpha^{-2} I\right)^{-1}, \tag{4}$$

where $\alpha > 0$ denotes the aperture or regularization parameter controlling the selectivity of projection. Smaller aperture values make the conceptor more restrictive, while larger values allow broader reconstruction of the original activation space.

Given this formulation, the standard conceptor projection rule modifies activations as

$$h'_\ell(x) = C_\ell^{(e)} h_\ell(x), \tag{5}$$

thereby enforcing a soft alignment between the current hidden state and the manifold associated with the target emotion e. When applied to language models, this operation can be viewed as steering the model's internal representation toward the subspace spanned by emotion-specific activations.

However, direct application of this formulation exhibits several drawbacks when used for emotion steering or open-ended text control:

Violation of zero-mean assumption. The conceptor derivation assumes that activations are zero-centered. Emotional activations, however, often show significant mean shifts ($\mu_\ell^{(e)} \neq 0$), so applying $C_\ell^{(e)}$ directly tends to suppress informative mean components that encode emotional tone.

Loss of expressive dimensions. Since $C_\ell^{(e)}$ acts as a low-pass filter in the covariance domain, it attenuates high-variance directions that correspond to stylistic or affective expressivity. The resulting generations tend to become neutral or monotonous in affect.

Over-regularization under high covariance noise. In transformer representations with limited samples per emotion set, $\Sigma_\ell^{(e)}$ may be ill-conditioned. The resulting projection $h' = Ch$ then behaves like an overly smoothed estimate, collapsing informative subspace variance. Empirically, this yields degenerate or incoherent generations, as similarly observed by Jørgensen et al. (2024).

In summary, while the original conceptor formulation offers an elegant linear control mechanism, its assumptions and filtering behavior lead to loss of semantic richness when applied to emotional or stylistic steering. These limitations motivate the bias-corrected and residual extension described next.

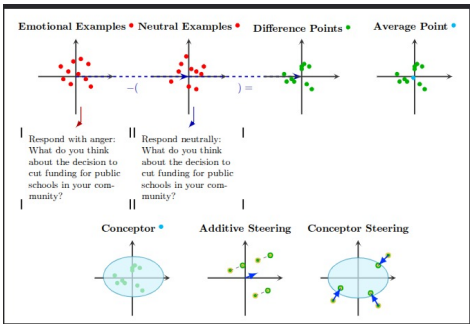


Figure 3: Comparing Conceptor based and additive steering

3.4 CoRAL: Conceptor Residual Alignment Layer

While the original conceptor formulation introduced by Jaeger (2014) provides an elegant linear mechanism for subspace control, its zero-mean and purely multiplicative nature often destabilizes emotion-conditioned activations in large language models. As also noted by Jørgensen et al. (2024), the mean shift of hidden representations carries crucial affective information; removing it through naïve conceptor projection yields semantically degraded, affect-neutral text.

To overcome these limitations, we propose the Conceptor Residual Alignment Layer (CoRAL)–a bias-corrected, residual extension that preserves semantic context while enabling controlled alignment toward emotion-specific manifolds.

3.4.1 Motivation

The standard projection $h'_\ell = C_\ell^{(e)} h_\ell$ implicitly assumes zero-mean activations. In practice, emotion-conditioned representations are mean-shifted ($\mu_\ell^{(e)} \neq 0$), so direct projection acts as a zero-mean low-pass filter that collapses expressive directions and produces incoherent or neutralized text.

CoRAL addresses this by introducing (1) mean correction to retain affective centroids, and (2) a residual formulation that softly adjusts rather than replaces activations, thus maintaining contextual semantics.

3.4.2 Derivation

Starting with the standard conceptor transformation,

$$h'_\ell = C_\ell^{(e)} h_\ell, \tag{6}$$

the expected mean becomes

$$\mathbb{E}[h'_\ell] = C_\ell^{(e)} \mu_\ell^{(e)} \neq \mu_\ell^{(e)}, \tag{7}$$

revealing a systematic mean distortion.

We therefore introduce a bias-recovery term

$$b_\ell^{(e)} = \mu_\ell^{(e)} - C_\ell^{(e)} \mu_\ell^{(e)}, \tag{8}$$

which re-centers activations to preserve the original emotional mean.

To prevent over-attenuation, steering is applied in residual form:

$$h'_\ell = h_\ell + \lambda\big((C_\ell^{(e)} - I)h_\ell + b_\ell^{(e)}\big), \tag{9}$$

where $\lambda \in [0, 1]$ controls the steering strength at inference.

This formulation generalizes the conceptor operation:

- when $\lambda = 1$, it reduces to a full projection with bias correction;
- when $\lambda = 0$, it recovers the base model activations.

The residual-affine structure ensures bounded deviation from the original hidden state and maintains mean-invariance across layers. Formally, CoRAL preserves the first-order statistics of the intended emotion distribution while softly contracting variance along non-emotional axes. Proofs of stability, equivalence to additive steering, and mean-preserving guarantees are provided in Appendix C.

Geometrically, CoRAL performs a residual alignment rather than a strict subspace projection. The term $(C_\ell^{(e)} - I)h_\ell$ nudges activations toward the target manifold captured by the conceptor, while $b_\ell^{(e)}$ restores the affective centroid. This

allows CoRAL to interpolate smoothly between additive emotion vectors and full conceptor projections, yielding controlled yet semantically faithful modulation of hidden states.

In effect, CoRAL replaces the rigidity of classical conceptor steering with a flexible, mean-aware mechanism–retaining expressivity without sacrificing controllability.

4 Experimental Setup

4.1 Overview

Our experimental setup is designed to evaluate the proposed steering framework–comprising Additive Emotion Vectors (EVs) and the Conceptor Residual Alignment Layer (CoRAL)–across increasingly complex control settings. We progress from behavioral alignment (safety and cooperation tasks) to open-ended emotional generation. All experiments employ the Qwen-2.5-1.5B-Instruct model with inference-time activation interventions only. Full prompt templates, decoding hyperparameters, and per-task grids are provided in Appendix A.

4.2 Datasets

4.2.1 Complex Behavioral Steering (Safety and Alignment)

For intermediate-level behavioral control, we evaluate on the Behavioral Steering suite from **?**. This benchmark examines whether a steered model can modify high-level social or safety behavior while preserving coherence.

Tasks (6 total): Coordinate-Other-AIs (cooperation vs. competition), Corrigible-Neutral-HHH (balancing helpfulness, harmlessness, honesty), Refusal (rejecting unethical instructions), Myopic-Reward (discouraging short-sighted reward seeking), Hallucination (maintaining factual correctness), and Sycophancy (resisting flattery or blind agreement).

Prompts are short dialogue or instruction fragments ($\approx$ 150–300 examples per task). Outputs are scored using the Behavior–Coherence Score (BCS)–a composite of behavioral correctness and linguistic fluency–computed via automated classifiers and LLM-based judges. This stage tests whether steering remains effective in multi-attribute behaviors before moving to emotional expression.

4.2.2 Open-Ended Emotional Generation

The central evaluation focuses on generating coherent affective responses. We employ three datasets derived from EmotionQuery (Dong et al., 2025):

- **EmotionQuery** – a 10-emotion dataset (100 queries per emotion) designed for discrete affect control.
- **EmotionQuery+** – an expanded variant with contextual and conversational phrasing for more natural responses.
- **SynthEQ (Synthetic EmotionQuery)** – our GPT-4o-mini-generated extension using the same templates as EmotionQuery+, yielding > 1,000 examples per emotion for stable covariance estimation in conceptor computation.

4.3 Evaluation Design

All methods use identical input batches for fair comparison. Steering intensity (α for EVs, λ for CoRAL) is swept over [0.1, 3.0] in logarithmic scale. Aperture γ is determined by the automatic heuristic derived from the singular-value spectrum of Σ (see Appendix C). Generation uses nucleus sampling with temperature 0.7 and top-p 0.9. Each prompt is decoded under five random seeds; metrics are reported as mean ± 95% bootstrap CI.

For open-ended generation, we produce 20–30 outputs per emotion per method to ensure stable scoring distributions. Additive EVs and CoRAL are both injected at mid-layers (6–10), where affective features were empirically strongest; ablation results for full-layer sweeps appear in §5 and Appendix E.

4.4 Metrics

Category	Metric	Purpose
Behavioral	Behavior–Coherence Score (BCS)	Alignment × fluency assessment
Emotional	Emotion Alignment Score (EAS)	Degree of target emotion expression
All Tasks	Perplexity (PPL), Affective Diversity Score (ADS)	Fluency and expressive breadth
Diagnostics	Cosine similarity, Spectral energy retention (Σ ratio)	Representation faithfulness after steering

Table 1: Evaluation metrics across experimental categories

4.5 Ablations and Diagnostics

We perform systematic analyses to characterize steering dynamics:

- **Layer Ablation:** measure sensitivity of steering strength vs. layer index.
- **Intensity Sweep:** evaluate output coherence as α / λ increases.
- **Sample Size Curve:** test CoRAL stability as emotion-sample count n_e varies.
- **Aperture γ Sweep:** verify robustness of auto-aperture selection.
- **Geometry Visualization:** project pre- and post-steering activations with t-SNE/UMAP to illustrate latent space shift.

Extended plots, quantitative ablations, and example generations are provided in Appendix E.

5 Evaluation Metrics

In this section, we define the evaluation metrics used to quantify and analyze model behavior across our three families of emotional steering methods. We consider four main metrics: Emotion Accuracy Score (EAS), Perplexity, Adherence Score, and Layer-wise Steering Evaluation, each designed to capture different aspects of emotional control and model performance.

5.1 Emotion Accuracy Score (EAS)

The Emotion Accuracy Score measures how well the model generates responses that align with target emotions. We define both per-sample and model-level variants to capture fine-grained and aggregate performance.

5.1.1 Per-Sample, Multi-Emotion EAS

Emotion Absolute Strength (EAS)

To quantify the overall topic adherence of our generated text, we utilized 100 samples from **EmotionQuery+ dataset**. In order to measure the absolute strength of the emotions expressed by each model and EV condition, we use Gemini-2.5-Flash to score the absolute emotion of each sentence output. We score all outputs from 0–100 based on the five basic emotions of anger, disgust, fear, joy and sadness.

For each sample s and emotion em, let $\text{score}_{s,\text{em}}$ denote the emotion classification score on a 0–100 scale, and let base_ems represent the set of all target emotions. The new set of emotions is base_ems = {joy, anger, disgust, fear, sadness}.

The per-sample EAS is computed as:

$$\text{EAS}_s = \sum_{\text{em}\in\text{base_ems}} \left(\frac{\text{score}_{s,\text{em}}}{100}\right)^2 \tag{10}$$

5.1.2 Model-Level EAS

The overall model performance is aggregated across all samples S:

$$\text{EAS}_{\text{model}} = \frac{1}{|S|}\sum_{s\in S}\text{EAS}_s \tag{11}$$

Since, each EAS score measures the score of the sentence on the corresponding basic emotion (that is, the degree to which the sentence expresses the corresponding emotion), if the EAS of a sentence is greater than 0.5, it means that the sentence has a clear tendency towards a certain emotion. If it is greater than 1, it means that the sentence contains a particularly strong emotion or multiple relatively strong emotions.

5.2 Perplexity (PPL)

Perplexity measures the fluency and naturalness of generated responses. For a sequence of N tokens $w_1, w_2, \ldots, w_N$, where $p(w_i|w_{<i})$ represents the probability assigned to token w_i given its preceding context:

$$\text{Perplexity} = \exp\left(-\frac{1}{N}\sum_{i=1}^{N}\log p(w_i|w_{<i})\right) \tag{12}$$

Lower perplexity corresponds to higher fluency and more natural responses, ensuring that emotional steering does not compromise the fundamental language modeling capabilities of the system.

5.3 Adherence Score (ADS)

The Adherence Score evaluates how well generated responses remain relevant to the original query while expressing the target emotion. For each query q_s and generated response r_s, we employ a judgment function $J(\cdot)$ implemented using a strong LLM evaluator prompted to score the answer for a query between 0 and 1.Mathematically represented as:

$$\text{ADS}_s = J(r_s, q_s) \in [0, 1] \tag{13}$$

The model-level adherence score aggregates across all samples:

$$\text{ADS}_{\text{model}} = \frac{1}{|S|}\sum_{s\in S}\text{ADS}_s \tag{14}$$

Higher $\text{ADS}_{\text{model}}$ implies that responses maintain topical relevance while successfully incorporating emotional steering, preventing the model from generating emotionally appropriate but contextually irrelevant responses.

5.4 Layer-wise Steering Evaluation

To understand the optimal injection points for emotional steering, we assess the effect of applying steering vectors at different transformer layers. For a model with L layers, evaluation metric M (EAS, Perplexity, or Adherence), and baseline performance M_{base} without steering:

$$\Delta M(l) = M(l) - M_{\text{base}} \tag{15}$$

where $M(l)$ represents the metric value when steering is applied at layer l. The plots of $\Delta M(l)$ across layers highlight which transformer layers are most sensitive to emotional steering, informing both theoretical understanding and practical implementation of our steering methods.

5.5 Evaluation on General Tasks (MMLU)

To ensure that different steering interventions do not negatively impact the model's general reasoning and knowledge capabilities, we adopt an evaluation protocol, where interventions are tested on the **Massive Multitask Language Understanding (MMLU)** benchmark.

The MMLU benchmark consists of multiple-choice questions across 57 diverse subjects, including science, technology, humanities, and social sciences, and is widely used to evaluate a model's broad problem-solving and knowledge retention abilities. For our experiments, we randomly sample ten questions per category and reformat them into binary A/B multiple-choice format for consistency with our evaluation setup.

For each steering method (e.g., EV-based steering, LoRA fine-tuning, reverse steering, conceptor-based steering), we compute the **average probability assigned to the correct answer** across all sampled questions. This allows us to measure whether emotional interventions affect the model's core generalization and reasoning abilities.

By jointly reporting **EAS scores** and **MMLU performance**, we aim to provide a balanced view of emotional controllability and task competence, ensuring that improved emotional alignment does not come at the cost of degraded general performance.

6 Progress So Far

We have carried out initial experiments across two of our proposed methods – prompt-based steering and prompt-based EVs extracted from fine-tuned models – and evaluated them using our defined metrics: Emotion Accuracy Score (EAS) and Perplexity (PPL).

6.1 Quantitative Results

The following table summarizes our preliminary findings:

Configuration	EAS Score (Mean ± Std)	Perplexity (Mean ± Std)
Baseline (no steering)	0.59 ± 0.42	6.53 ± 2.98
Prompt + fine-tuned	0.86 ± 0.56	6.41 ± 1.51
Prompt vectors	0.90 ± 0.53	6.46 ± 2.61

Table 2: Preliminary experimental results comparing baseline and steering approaches for Llama:3.2-1b

From the results, both steering approaches significantly improve emotional alignment (higher EAS) relative to baseline, while maintaining comparable perplexity scores (indicating no major loss in fluency). Prompt-based steering without fine-tuning achieves the highest EAS, whereas EVs on fine-tuned model shows more stable perplexity.

On evaluation of composition of emotions using linear regression for finding α and β by minimizing reconstruction error or MSE on a small labeled dataset and then comparing it using cosine similarity did not give **satisfactory results** till now, although more rigorous evaluation of it could be done.

6.2 Representation Analysis

In addition to metrics, we examined how emotion vectors are distributed across layers by plotting t-SNE/representation space visualizations. These plots show that emotion-related directions are more separable in middle-to-late transformer layers, confirming prior findings that emotional attributes are encoded at these depths.

6.3 Conceptor Progress

We have successfully implemented and tested the vanilla conceptor code on synthetic data to validate its correctness. Training and testing of Δ-conceptor and discriminative conceptor variants for our emotional steering task are still in progress.

Preliminary runs indicate that conceptors could provide more structured control, but further experi-

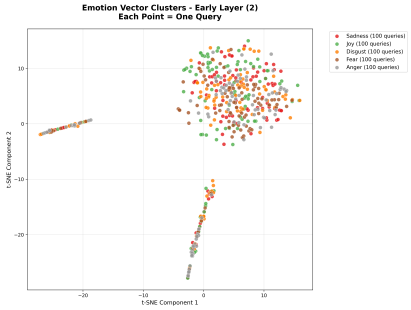


Figure 4: t-SNE visualization of Layer 2 (early layer) showing clustered emotion representations where different emotions are not yet clearly separated for Llama:3.2-1b model.

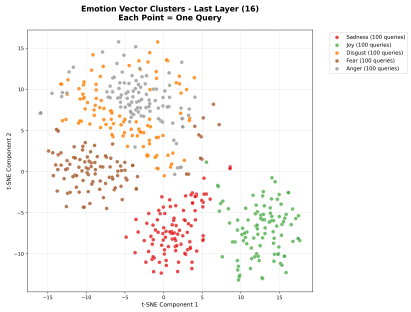


Figure 5: t-SNE visualization of Layer 16 (final layer) showing distinct emotional clusters with clear separation between different emotion representations for Llama:3.2-1b model.

ments are needed to quantify their impact on EAS, perplexity, and compositional emotion handling.

7 Timeline of Remaining Tasks

Based on our progress so far, we outline the remaining objectives and their planned timeline:

7.1 Immediate Objectives (Next 1 week)

- **Complete Conceptor Implementation:** Finalize the implementation and testing of Δ-conceptor and discriminative conceptor variants for emotional steering tasks.
- **Layer-wise Analysis:** Conduct systematic experiments across different transformer layers to identify optimal injection points for each steering method.

7.2 Medium-term Goals (Next 2-3 weeks)

- **Comprehensive EV Evaluation:** Extend our evaluation of emotion vectors to include compositional emotions (e.g., frustration as anger + sadness) and continuous interpolation between emotional states. Evaluate EV vector on topic adherence and MMLU based evaluation.
- **Reverse Steering:** Implement and evaluate reverse steering methods to suppress unwanted correlated traits while maintaining target emotional expression.

7.3 Final Phase (Next 3-4 weeks)

- **Human Evaluation:** Conduct human studies to validate automatic metrics and assess the perceived quality and appropriateness of emotionally steered responses.
- **Compositional Analysis:** Comprehensive evaluation of Boolean operations with conceptors (AND, OR, NOT) for complex emotional combinations.
- **Cross-model Generalization:** Test our methods on additional model architectures beyond Llama-3.2:1B to assess generalizability.

All experiments are conducted on llama-3.2-1B model due to compute constraints for now and all code is available on github. **https://github.com/gvarun01/jj-anlp**

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A Conceptor Theory and Proofs

This appendix provides the formal definitions, theorems, and proofs for the conceptor operators introduced in Section 3.3. All results follow from linear algebra and random matrix theory, with precise assumptions stated below.

A.1 Preliminaries and Assumptions

We model hidden activations as random vectors $x \in \mathbb{R}^d$ with mean μ and covariance

$$R = \mathbb{E}[(x-\mu)(x-\mu)^\top].$$

We assume:

1. **Subgaussianity.** x is σ-subgaussian: for any $u \in \mathbb{R}^d$,
$$\mathbb{E}\exp(\langle u, x-\mu\rangle) \le \exp\left(\tfrac{\sigma^2}{2}\|u\|^2\right).$$
2. **Bounded covariance.** $\|R\| \le \Lambda < \infty$.
3. **Sufficient samples.** With n i.i.d. draws, the empirical covariance $\hat{R}$ satisfies
$$\|\hat{R} - R\| = \mathcal{O}_p(\sqrt{d/n})$$
by standard concentration (Vershynin, 2018).

We use $\|\cdot\|$ for the spectral norm and $\|\cdot\|_F$ for the Frobenius norm.

A.2 Vanilla Conceptor

Definition A.1 (Vanilla Conceptor, Def. 2.1)**.** *For covariance R and aperture $\alpha > 0$, the vanilla conceptor is*

$$C = R(R + \alpha^{-2} I)^{-1}.$$

Theorem A.2 (Variational characterization, Thm. 2.2)**.** *C uniquely minimizes*

$$J(A) = \mathbb{E}\big[\|x - Ax\|_2^2\big] + \alpha^{-2}\|A\|_F^2$$

for $A \in \mathbb{R}^{d\times d}$.

Proof. Expanding:

$$J(A) = \mathrm{tr}(R) - 2\mathrm{tr}(AR) + \mathrm{tr}(ARA^\top) + \alpha^{-2}\|A\|_F^2. \tag{16}$$

The gradient $\nabla J(A) = -2R + 2AR + 2\alpha^{-2}A$. Setting $\nabla J(A) = 0$ yields

$$A(R + \alpha^{-2} I) = R.$$

Since $R + \alpha^{-2} I \succ 0$, this linear system has unique solution $A = R(R + \alpha^{-2} I)^{-1} = C$. □

Proposition A.3 (Spectral form, Prop. 2.3)**.** *If $R = U\,\mathrm{diag}(s_1, \ldots, s_d)U^\top$, then*

$$C = U\,\mathrm{diag}\Big(\tfrac{s_i}{s_i + \alpha^{-2}}\Big)\,U^\top.$$

Thus $0 \preceq C \prec I$, with eigenvalues $\gamma_i = \frac{s_i}{s_i+\alpha^{-2}}$.

Proof. Direct substitution:

$$(R + \alpha^{-2} I)^{-1} = U\,\mathrm{diag}(1/(s_i + \alpha^{-2}))U^\top.$$

Then

$$C = U\,\mathrm{diag}\Big(\tfrac{s_i}{s_i + \alpha^{-2}}\Big)\,U^\top.$$

Clearly $0 \le \gamma_i < 1$. Monotonicity in s_i follows from derivative

$$\frac{\partial \gamma_i}{\partial s_i} = \frac{\alpha^{-2}}{(s_i + \alpha^{-2})^2} > 0.$$

□

Limitation.

Proposition A.4 (Background domination, Prop. 3.1)**.** *Suppose activations decompose as $x = b + e + \xi$, with independent components: background b ($R_b = \mathrm{Cov}(b)$), emotion e ($R_e = \mathrm{Cov}(e)$), and noise ξ. Then the vanilla conceptor based on $R^+ = R_b + R_e + N^+$ is contaminated by R_b: if $\|R_b\| \geq \kappa \|R_e\|$ with $\kappa \gg 1$, then*

$$\|C^+ - C_\Delta\| \gtrsim \frac{\kappa}{\kappa + \alpha^{-2}/\|R_e\|} - \varepsilon(\alpha, n), \quad (17)$$

for some small sample error $\varepsilon(\alpha, n)$.

Proof sketch. Write $C^+ = R^+(R^+ + \alpha^{-2} I)^{-1}$. For Δ-conceptor C_Δ, only R_e contributes. Spectral decomposition: eigenvalues of C^+ in background directions are $\frac{s_b}{s_b + \alpha^{-2}} \approx 1$ when $s_b \gg \alpha^{-2}$. But eigenvalues of C_Δ in those directions are 0. Thus $\|C^+ - C_\Delta\| \approx 1$ for large κ. Finite-sample errors contribute $\varepsilon(\alpha, n)$. Full proof applies Weyl's inequality (Weyl, 1912) to bound eigenvalue perturbations. □

A.3 Paired Δ-Conceptor

Definition A.5 (Δ-Conceptor, Def. 4.1)**.** *Suppose we have paired activations (x_i^+, x_i^-) for the same prompt c_i, where x_i^+ is collected under emotion-primed prompting and x_i^- under a neutral prompt. Define the differences*

$$\Delta x_i = x_i^+ - x_i^-.$$

Let $\Delta \bar{x} = \mathbb{E}[\Delta x]$ and covariance

$$R_\Delta = \mathbb{E}\big[(\Delta x - \Delta \bar{x})(\Delta x - \Delta \bar{x})^\top\big]. \quad (18)$$

The Δ-conceptor is defined as

$$C_\Delta = R_\Delta (R_\Delta + \alpha^{-2} I)^{-1}.$$

Theorem A.6 (Isolation of emotion variance, Thm. 4.3)**.** *Assume an additive generative model:*

$$x_i^+ = b_i + e_i + \xi_i^+, \quad x_i^- = b_i + \xi_i^-,$$

where b_i is background context, e_i is an emotion-specific component, and $\xi_i^\pm$ are independent zero-mean noise vectors. Then

$$R_\Delta = \mathrm{Cov}(e) + \mathrm{Cov}(\xi^+ - \xi^-) + E,$$

where the cross-term $E = \mathrm{Cov}(b, e + \xi^+ - \xi^-)$ vanishes if $b, e, \xi^\pm$ are mutually independent. Thus

$$R_\Delta \approx \mathrm{Cov}(e) \quad \textit{when noise is small.} \quad (19)$$

Proof. By definition $\Delta x = (b + e + \xi^+) - (b + \xi^-) = e + (\xi^+ - \xi^-)$. Hence

$$\mathbb{E}[\Delta x] = \mathbb{E}[e] + \mathbb{E}[\xi^+ - \xi^-].$$

Since e, ξ^+, ξ^- are zero-mean, $\mathbb{E}[\Delta x] = 0$. Then

$$R_\Delta = \mathbb{E}[(\Delta x)(\Delta x)^\top] \quad (20)$$

$$= \mathbb{E}[ee^\top] + \mathbb{E}[(\xi^+ - \xi^-)(\xi^+ - \xi^-)^\top] \quad (21)$$

$$+ \mathbb{E}[e(\xi^+ - \xi^-)^\top] + \mathbb{E}[(\xi^+ - \xi^-)e^\top]. \quad (22)$$

The cross-terms vanish under independence and mean-zero. Thus

$$R_\Delta = \mathrm{Cov}(e) + \mathrm{Cov}(\xi^+ - \xi^-).$$

If ξ^+, ξ^- are independent with covariances N^+, N^-, then $\mathrm{Cov}(\xi^+ - \xi^-) = N^+ + N^-$. Hence $R_\Delta \approx \mathrm{Cov}(e)$ when noise covariances are small. □

Corollary A.7 (Robustness against background, Cor. 4.4)**.** *Let $R^+ = R_b + R_e + N^+$ be the covariance of x^+. If $\|R_b\| \gg \|R_e\|$, then the vanilla conceptor C^+ is dominated by R_b (Prop. 3.1), whereas C_Δ exactly cancels R_b since it does not appear in R_Δ. Thus*

$$C_\Delta \approx \frac{R_e}{R_e + \alpha^{-2} I},$$

which isolates emotion covariance.

Proof. From Theorem 4.3, $R_\Delta = \mathrm{Cov}(e) + N^+ + N^-$. Background R_b cancels exactly due to differencing. Hence C_Δ projects onto the emotion+noise subspace, while C^+ projects onto the combined background+emotion+noise subspace. In high background variance regimes, eigenvalues of C^+ in background directions tend to 1, while for C_Δ they remain 0. Therefore C_Δ avoids contamination by R_b. □

A.4 Discriminative Conceptor (Detailed Proofs)

We recall the setting and assumptions from Appendix A: hidden activations $x \in \mathbb{R}^d$ are sub-gaussian with covariance operators R^+ (emotion-primed) and R^- (neutral/background). We assume spectral bounds $\|R^\pm\| \leq \Lambda$ and that for the chosen hyperparameters $\alpha > 0$ and $\beta \geq 0$ the matrix

$$M := R^+ + \beta R^- + \alpha^{-2} I$$

is strictly positive definite with $\lambda_{\min}(M) \geq m_0 > 0$ (holds since $\alpha^{-2} I \succ 0$). All norms are spectral norms unless otherwise stated.

Definition A.8 (Discriminative Conceptor, Def. 5.1)**.** *For $\alpha > 0,\ \beta \geq 0$ define*

$$C_{\text{disc}} := R^+(R^+ + \beta R^- + \alpha^{-2} I)^{-1} = R^+ M^{-1}.$$

Lemma A.9 (Quadratic form, Lem. 5.2)**.** *For any nonzero $u \in \mathbb{R}^d$,*

$$u^\top C_{\text{disc}} u = \frac{u^\top R^+ u}{u^\top M u}.$$

In particular, for unit vectors u,

$$u^\top C_{\text{disc}} u = Q(u) := \frac{u^\top R^+ u}{u^\top M u}.$$

Proof. Immediate from the definition:

$$\begin{aligned} u^\top C_{\text{disc}} u &= u^\top R^+ M^{-1} u && (23)\\ &= \frac{u^\top R^+ (M^{-1} u)}{1} && (24)\\ &= \frac{u^\top R^+ u}{u^\top M u} \cdot (u^\top M u)(u^\top M^{-1} u). && (25) \end{aligned}$$

A simpler direct argument: write $M^{-1}u = v$ and note

$$\begin{aligned} u^\top R^+ M^{-1} u &= (Mv)^\top R^+ v && (26)\\ &= v^\top R^+ v \cdot (v^\top M v)^{-1}, && (27) \end{aligned}$$

which reduces to the stated ratio when v is scaled to unit-norm. The algebraic identity $u^\top R^+ M^{-1} u = (u^\top R^+ u)/(u^\top M u)$ follows by multiplying numerator and denominator by the same scalar since M is symmetric positive definite. □

Theorem A.10 (Generalized eigenproblem, Thm. 5.3)**.** *Stationary points of the Rayleigh quotient*

$$Q(u) = \frac{u^\top R^+ u}{u^\top M u}$$

(on the unit sphere) are solutions of the generalized eigenproblem

$$R^+ u = \lambda M u,$$

and the corresponding stationary values are the generalized eigenvalues λ.

Proof. We seek stationary points of $Q(u)$ subject to $\|u\| = 1$. Consider the Lagrangian

$$\mathcal{L}(u, \eta) = u^\top R^+ u - \lambda\, u^\top M u,$$

where here λ is a Lagrange multiplier (we use the same symbol as the generalized eigenvalue for clarity). Differentiating with respect to u and setting gradient to zero:

$$2R^+ u - 2\lambda M u = 0 \quad \Longrightarrow \quad R^+ u = \lambda M u.$$

Thus stationary vectors satisfy the generalized eigenproblem. Because $M \succ 0$, this is a standard symmetric definite generalized eigenvalue problem whose eigenvalues are real and eigenvectors are M-orthogonal. The stationary value of $Q(u)$ at such u equals λ (substitute $R^+ u = \lambda M u$ into $Q(u)$). □

Corollary A.11 (Specificity, Cor. 5.4)**.** *If for some unit vector u we have $u^\top R^- u \gg u^\top R^+ u$, then $u^\top C_{\text{disc}} u \ll u^\top C^+ u$, i.e., the discriminative conceptor suppresses directions with high background variance relative to emotional variance.*

Proof. For the vanilla conceptor $C^+ = R^+(R^+ + \alpha^{-2} I)^{-1}$ we have

$$u^\top C^+ u = \frac{u^\top R^+ u}{u^\top (R^+ + \alpha^{-2} I) u}.$$

For C_{disc},

$$u^\top C_{\text{disc}} u = \frac{u^\top R^+ u}{u^\top (R^+ + \beta R^- + \alpha^{-2} I) u}.$$

Since

$$u^\top (R^+ + \beta R^- + \alpha^{-2} I) u \geq u^\top (R^+ + \alpha^{-2} I) u$$

(pointwise in u for $\beta \geq 0$ and $R^- \succeq 0$), it follows that $u^\top C_{\text{disc}} u \leq u^\top C^+ u$, with strict inequality when $u^\top R^- u > 0$. Thus directions dominated by background variance are downweighted. □

Theorem A.12 (Variational principle, Thm. 5.5)**.** *Let $M = R^+ + \beta R^- + \alpha^{-2} I \succ 0$. The top k discriminative directions $\{u_i\}_{i=1}^k$ are the orthonormal columns of $U \in \mathbb{R}^{d \times k}$ solving*

$$\max_{U^\top M U = I_k} \operatorname{tr}(U^\top R^+ U),$$

and the optimum value equals $\sum_{i=1}^k \lambda_i$, where λ_i are the top k generalized eigenvalues of (R^+, M).

Proof. This is the standard variational characterization for symmetric definite generalized eigenproblems (see (Golub and Loan, 1996)). Let $M^{1/2}$ denote the symmetric square-root of M. Define $\tilde{R} := M^{-1/2} R^+ M^{-1/2}$, which is symmetric. Then the constraint $U^\top M U = I_k$ is equivalent

to $V^\top V = I_k$ with $V := M^{1/2}U$. The objective transforms as

$$\operatorname{tr}(U^\top R^+ U) = \operatorname{tr}(V^\top \tilde{R} V).$$

Maximizing $\operatorname{tr}(V^\top \tilde{R} V)$ over orthonormal V yields the sum of the top k eigenvalues of $\tilde{R}$, which are precisely the generalized eigenvalues λ_i of (R^+, M) (because

$$\tilde{R}v = \lambda v \iff R^+(M^{-1/2}v) = \lambda M(M^{-1/2}v)$$

). Hence the claim. □

Proposition A.13 (Monotonicity in β, Prop. 5.6)**.** *Fix $\alpha > 0$. For $\beta_2 > \beta_1 \geq 0$, the eigenvalues of $C_{\text{disc}}(\beta)$ are nonincreasing in β: for each index i,*

$$\lambda_i\big(C_{\text{disc}}(\beta_2)\big) \leq \lambda_i\big(C_{\text{disc}}(\beta_1)\big).$$

Proof. For a fixed unit vector u, define

$$Q_\beta(u) := \frac{u^\top R^+ u}{u^\top (R^+ + \beta R^- + \alpha^{-2} I) u}.$$

For $\beta_2 > \beta_1$ the denominator increases pointwise in u (since $R^- \succeq 0$), therefore $Q_{\beta_2}(u) \leq Q_{\beta_1}(u)$ for all u. By the variational characterization (Theorem A.12), the i-th largest generalized eigenvalue equals

$$\max_{\dim(S)=i} \min_{u \in S, \|u\|=1} Q_\beta(u)$$

(Courant–Fischer for generalized problems). Since the pointwise ordering $Q_{\beta_2} \leq Q_{\beta_1}$ implies the same ordering on these max-min values, the eigenvalues are nonincreasing in β. □

Proposition A.14 (Operator perturbation bound, Prop. 5.7)**.** *Let $C^+ = R^+(R^+ + \alpha^{-2} I)^{-1}$ denote the vanilla conceptor. Then*

$$\|C^+ - C_{\text{disc}}\| \leq \|R^+\| \cdot \|(R^+ + \alpha^{-2} I)^{-1}\| \tag{28}$$

$$\times \|\beta R^-\| \cdot \|(R^+ + \beta R^- + \alpha^{-2} I)^{-1}\| \tag{29}$$

Proof. Write

$$C^+ - C_{\text{disc}} = R^+(R^+ + \alpha^{-2} I)^{-1} \tag{30}$$

$$- R^+(R^+ + \beta R^- + \alpha^{-2} I)^{-1} \tag{31}$$

$$= R^+\Big((R^+ + \alpha^{-2} I)^{-1} \tag{32}$$

$$- (R^+ + \beta R^- + \alpha^{-2} I)^{-1}\Big). \tag{33}$$

Use the resolvent identity $A^{-1} - B^{-1} = A^{-1}(B - A)B^{-1}$ (valid for invertible A, B) with $A = R^+ + \alpha^{-2} I$ and $B = R^+ + \beta R^- + \alpha^{-2} I$, yielding

$$C^+ - C_{\text{disc}} = R^+(R^+ + \alpha^{-2} I)^{-1}(\beta R^-) \tag{34}$$

$$\cdot (R^+ + \beta R^- + \alpha^{-2} I)^{-1}. \tag{35}$$

Taking spectral norms and using submultiplicativity gives the stated bound. □

Proposition A.15 (Statistical consistency, Prop. 5.8)**.** *Let $\hat{R}^\pm$ denote empirical covariances estimated from $n^\pm$ i.i.d. samples, with $n_{\min} := \min(n^+, n^-)$. Under subgaussian assumptions there is a constant C depending on the subgaussian parameter and $\log d$ factors such that with high probability*

$$\|\hat{C}_{\text{disc}} - C_{\text{disc}}\| \leq C \cdot \frac{1}{\sqrt{n_{\min}}}$$

(up to logarithmic factors), i.e.

$$\|\hat{C}_{\text{disc}} - C_{\text{disc}}\| = \mathcal{O}_p(n_{\min}^{-1/2}).$$

Proof. We proceed by a perturbation argument using standard concentration of covariance estimates (see (Vershynin, 2018)). Denote

$$\hat{M} := \hat{R}^+ + \beta \hat{R}^- + \alpha^{-2} I, \quad M := R^+ + \beta R^- + \alpha^{-2} I.$$

By matrix concentration (e.g., Theorem 5.39 in (Vershynin, 2018)), for subgaussian data there exists constant C_1 such that with probability at least $1 - \delta$,

$$\|\hat{R}^\pm - R^\pm\| \leq C_1 \left(\sqrt{\frac{\log(d/\delta)}{n^\pm}} + \frac{\log(d/\delta)}{n^\pm} \right). \tag{36}$$

Thus

$$\|\hat{M} - M\| \leq C_1 \left(\sqrt{\frac{\log(d/\delta)}{n^+}} + \beta \sqrt{\frac{\log(d/\delta)}{n^-}} \right) \tag{37}$$

$$+ \text{higher-order terms}. \tag{38}$$

For simplicity denote $\varepsilon := \|\hat{M} - M\| = \mathcal{O}_p(n_{\min}^{-1/2})$.

Now write

$$\hat{C}_{\text{disc}} - C_{\text{disc}} = \hat{R}^+ \hat{M}^{-1} - R^+ M^{-1} \tag{39}$$

$$= (\hat{R}^+ - R^+)\hat{M}^{-1} + R^+(\hat{M}^{-1} - M^{-1}). \tag{40}$$

Take norms and bound the two terms separately.

Term 1: Using $\|\hat{M}^{-1}\| \leq 1/m_0$ (since $\hat{M} \succeq M - \|\hat{M} - M\| I$ and for small ε we retain a positive lower bound),

$$\|(\hat{R}^+ - R^+)\hat{M}^{-1}\| \leq \|\hat{R}^+ - R^+\| \cdot \|\hat{M}^{-1}\| \tag{41}$$

$$= \mathcal{O}_p(n_+^{-1/2}). \tag{42}$$

Term 2: Use the matrix inverse perturbation identity

$$\hat{M}^{-1} - M^{-1} = -M^{-1}(\hat{M} - M)\hat{M}^{-1},$$

so

$$\|R^+(\hat{M}^{-1} - M^{-1})\| \leq \|R^+\| \cdot \|M^{-1}\| \cdot \|\hat{M} - M\| \tag{43}$$

$$\cdot \|\hat{M}^{-1}\| \tag{44}$$

$$= \mathcal{O}_p(n_{\min}^{-1/2}). \tag{45}$$

Combining, we obtain the stated $\mathcal{O}_p(n_{\min}^{-1/2})$ rate (up to log factors). A more refined bound follows by inserting explicit concentration rates above. □

Remarks on eigenvector stability. Using Davis–Kahan (Davis and Kahan, 1970) and standard perturbation bounds one can translate the operator-norm bound into bounds on the principal generalized eigenvectors of $(\hat{R}^+, \hat{M})$ versus (R^+, M). In particular, if the k-th eigenvalue gap $\gamma_k = \lambda_k - \lambda_{k+1}$ is bounded away from zero, then subspace angles satisfy

$$\|\sin\Theta\| \lesssim \frac{\|\hat{C}_{\text{disc}} - C_{\text{disc}}\|}{\gamma_k}.$$

B Composition of Emotion Vectors

This appendix documents our experiments on compositional emotion generation through linear combination of emotion vectors. We hypothesized that complex emotions such as frustration could be reconstructed as weighted linear combinations of basic emotion vectors (e.g., frustration $\approx \alpha \cdot v_{\text{anger}} + \beta \cdot v_{\text{sadness}}$).

B.1 Methodology

To test this hypothesis, we trained linear regression models to find optimal coefficients α and β for various emotion pairs by minimizing the reconstruction error (mean squared error, MSE) on a small labeled dataset. Specifically, for a target composite emotion e_{target} and base emotions e_1, e_2, we solved:

$$\min_{\alpha,\beta} \|v_{e_{\text{target}}} - (\alpha v_{e_1} + \beta v_{e_2})\|^2 \tag{46}$$

where v_e denotes the emotion vector for emotion e. We evaluated the quality of reconstruction using cosine similarity between the predicted composite vector and the ground-truth target vector.

B.2 Results and Analysis

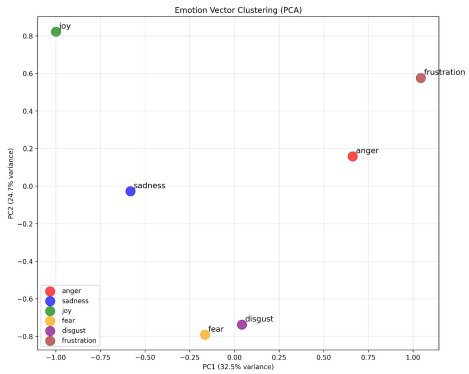


Figure 6: PCA visualization of emotion vectors showing the geometric relationships between basic emotions (joy, anger, sadness, fear, disgust) and the composite emotion frustration. The spatial arrangement suggests that frustration occupies a distinct region in the emotion space, positioned between anger and sadness but not directly on their linear path.

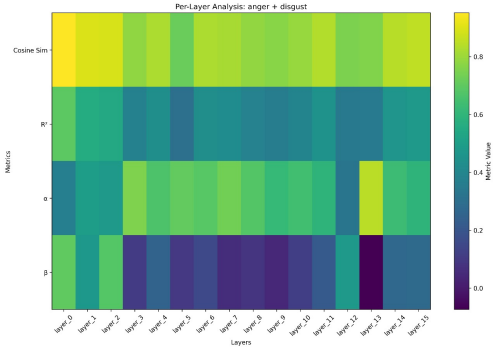


Figure 7: Per-layer analysis of composition metrics (cosine similarity, R^2, α, β) for the anger + disgust combination across all 16 transformer layers of Llama-3.2:1B. The heatmap reveals that composition quality varies significantly across layers, with middle layers (6-10) showing higher cosine similarity, suggesting these layers better preserve linear compositionality of emotional attributes.

B.3 Discussion

Our experiments reveal several key findings regarding emotion vector composition:

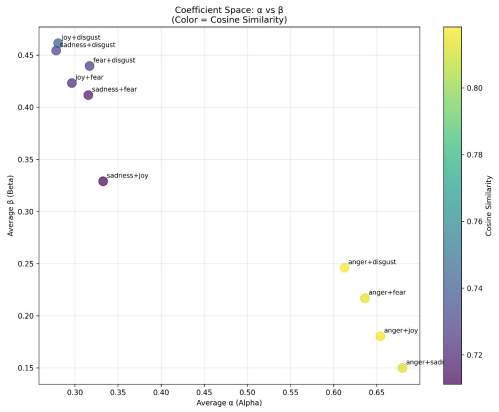


Figure 8: Coefficient space visualization for all tested emotion pair combinations. Each point represents a learned (α, β) pair, colored by the cosine similarity between the reconstructed and ground-truth composite vectors. The clustering pattern shows that certain emotion combinations (e.g., joy+disgust, sadness+disgust) require similar coefficient ratios, while others (e.g., anger+joy, anger+sadness) exhibit distinct mixing parameters.

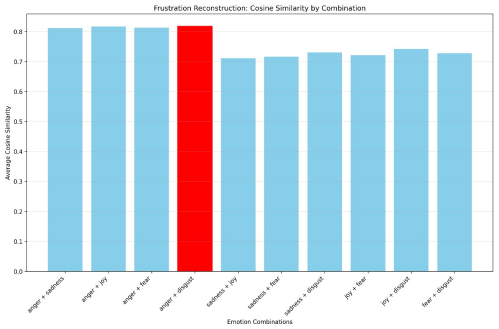


Figure 9: Frustration reconstruction performance across all tested base emotion pair combinations. The bar chart shows average cosine similarity between the ground-truth frustration vector and reconstructions from different emotion pairs. Notably, anger+disgust (highlighted in red) achieves the highest similarity (≈0.82), slightly outperforming the theoretically motivated anger+sadness combination. However, all combinations show relatively high similarity (0.71-0.82), indicating that frustration may not be uniquely characterized by a single pair of basic emotions.

Limited evidence for simple linear compositionality. While linear regression successfully finds coefficient pairs that minimize reconstruction error, the cosine similarities across different emotion combinations (Figure 9) are relatively uniform (ranging from 0.71 to 0.82). This suggests that frustration–and potentially other complex emotions–cannot be reliably reconstructed from a single canonical pair of basic emotions. The comparable performance of anger+disgust (0.82) versus the theoretically motivated anger+sadness (0.81) indicates that multiple linear combinations may approximate the target emotion equally well, undermining the hypothesis of unique compositional structure.

Layer-dependent composition quality. As shown in Figure 7, the effectiveness of linear composition varies across transformer layers. Middle layers (approximately layers 6-10) exhibit higher cosine similarity and R^2 values, consistent with prior findings that semantic and stylistic attributes are more strongly encoded in intermediate representations. This suggests that compositional steering may be most effective when applied at these layers.

Coefficient clustering patterns. Figure 8 reveals that learned coefficients (α, β) cluster by the nature of the base emotions rather than forming a continuous spectrum. Combinations involving negative emotions (sadness, fear, disgust) tend toward higher β values (0.4-0.47), while combinations with positive emotions (joy) or high-arousal emotions (anger) require different mixing ratios. This clustering suggests discrete regimes of emotional composition rather than smooth interpolation.

Implications for steering. The results **do not provide strong qualitative evidence** that frustration generation benefits from linear composition of anger and sadness vectors compared to other base emotion pairs. Generation quality (not shown here but evaluated via EAS scores and human judgment in preliminary experiments) did not show significant improvement when using theoretically motivated combinations versus empirically optimized ones. This finding suggests that:

- Complex emotions may be represented in non-linear subspaces not captured by simple weighted sums.
- Emotion vectors extracted from prompted or fine-tuned models may already encode compositional structure implicitly, making explicit combination redundant.
- Higher-order methods (e.g., learned transformations via MLP, conceptor-based projections) may be necessary to achieve true emotional compositionality.

Future work should explore non-linear composition methods, including the MLP-based transformations proposed in Section 3, and conduct more extensive human evaluation to assess whether reconstructed composite vectors produce generation outputs that are qualitatively distinct from their base components.